FIT2179 Data Visualisation

Course Notes

DOT MAPS

13 September 2026

DOT MAPS

A dot map uses point marks to show a quantitative attribute. Dot maps show raw totals for a phenomenon that is not uniformly distributed. Each dot represents a certain quantity of that phenomenon. A simple dot map might show the location of meteor impacts, with each dot representing one impact. More often, though, a dot represents many items. On the map below, each dot represents 1,000 cows. Representing every individual cow with a dot would not be feasible, as the result would be a black blob.



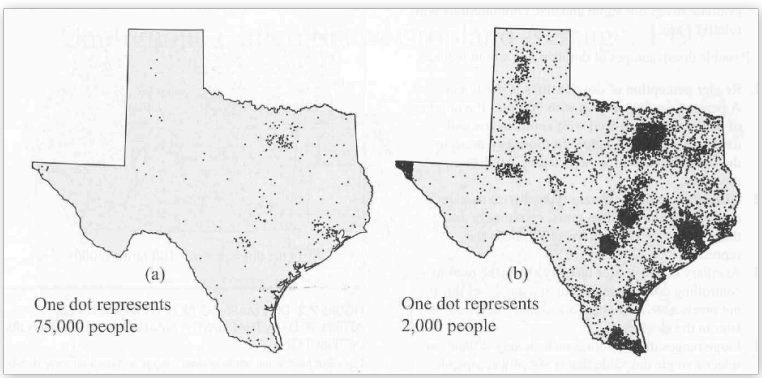
CONSTRUCTING A DOT MAP

Construction usually starts with a quantitative attribute per geographic area. The unit value per dot (1000 in the cow map) and the dot size are chosen. The unit value should be easy to add up, such as 100, 250, 5000, etc. Each attribute value is then divided by the unit value and the resulting number of dots is placed within the geographic area. Where the dots fall within the area can be controlled by defining exclusion areas, which ensure that dots are only placed where the phenomenon actually occurs. On the cow map, dot placement could be restricted to rural areas.

Dot maps covering an entire continent or the whole world need an equal-area projection. Using the Mercator projection for meteor impacts would be wrong, because the Mercator projection enlarges areas near the poles, giving the impression that impacts are rare at polar latitudes.

WHEN TO USE DOT MAPS

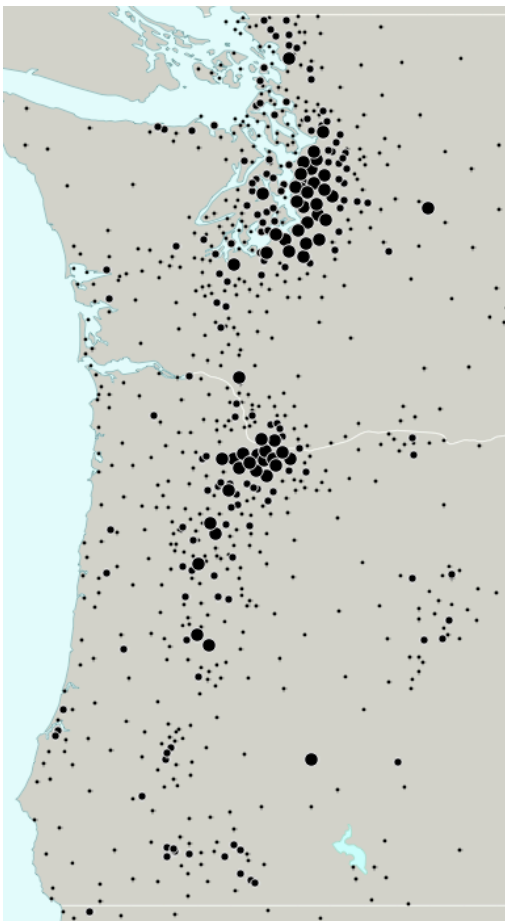
Dot maps do not work well if the mapped phenomenon is strongly concentrated in a few areas. The two maps of the population of Texas are not particularly successful. On the left, one dot represents a large number of people: the dot density is appropriate for cities, but most of the map is empty and the population pattern across most of Texas is lost. On the right, one dot represents a small number of people, which reveals interesting patterns in rural Texas, but the dots are far too dense in the cities. In such situations, another map type should be used, such as a choropleth map, bin map, proportional symbol map, or isocontour map.



Map from Slocum et al. 2022, Thematic Cartography and Geovisualization, ed. 4.

GRADUATED DOT MAPS

Graduated dot maps use several dot sizes. The map below, showing the population of the north-western United States, uses three sizes: a small dot represents 2,000 inhabitants, a medium dot 10,000, and a large dot 100,000. Graduated dot maps are rarely used, because no easy-to-use software exists for creating them.



Map by N. Arnold, Oregon State University.